NISM SERIES VIII · MARCH 2026 WORKBOOK · STUDY NOTES

Equity Derivatives

Futures, options and the maths of risk: learn to price, hedge and trade them the way an F&O desk does.

A derivative is a contract whose value comes from something else, like a share or the Nifty. It works a bit like booking a hotel room months ahead at a fixed rate: a futures contract fixes the price for both sides, while an option is like paying a small non-refundable booking fee for the right, but not the duty, to take the room. This certification is for people who sit at a broker's F&O terminal or sell F&O to clients. SEBI requires it because derivatives are leveraged. A small margin controls a large position, so the people dealing with investors must understand pricing, margins, settlement and risk before they handle client money.

📋 Exam pattern

100	120 min	60%	25% of the marks assigned to the question
Total marks	Duration	Pass mark	Negative marking

100 multiple choice questions · Numerical-heavy paper: payoffs, break-evens, fair futures price, hedge ratios, margins, MTM, corporate-action adjustments. Test centre workstations provide Microsoft Excel or LibreOffice for numericals. Applies to exams taken on or after April 20, 2026 (workbook version March 2026).

👥 Who needs it

- Associated persons working as approved users of a trading member on an equity derivatives exchange or the equity derivatives segment of a recognised stock exchange
- Sales personnel of trading members dealing in equity F&O
- Every trading member must appoint at least two approved users who have passed the SEBI-approved certification; only approved users may operate the derivatives trading terminal

🧭 Exam strategy

- Budget about 70 seconds per question: do a first pass of 75 minutes on everything you know, then 35 minutes on numericals, then 10 minutes to review.
- Chapters 3 and 4 carry 40 marks. If you are perfect there plus Ch 5, 6 and 7, you are already near the pass mark.
- Use the on-screen Excel or LibreOffice for e^x (EXP), powers and hedge ratios, and for continuous-compounding fair value questions.
- For any strategy question, quickly tabulate the payoff at three prices (below the low strike, between the strikes, above the high strike). Max profit, max loss and BEP become obvious.
- Watch the sign conventions: basis = spot − futures; BEP for a call adds the premium, for a put subtracts it; delta of a put is negative.
- Negative marking is 25%: guess only if you can eliminate at least one option (the expected value is then positive). Leave pure guesses blank.
- Read 'NOT', 'EXCEPT' and 'per lot' versus 'per share' carefully. Many numericals multiply by the lot size.
- Regulatory numbers change often. Trust the March 2026 workbook values (₹15–20 lakh lots, STT 0.05%/0.15%, Tuesday/Thursday expiries).
- For open-position questions, remember that proprietary positions are netted and client positions are grossed.



Chapter 1 · Weight: 10%

Basics of Derivatives

This chapter covers what a derivative is (a contract that takes its value from an underlying), the four building blocks (forwards, futures, options and swaps), how India's F&O market evolved, who the participants are, and the risks involved.

> **MAKE IT STICK**
> A 13th-century English monastery sold its wool up to 20 years in advance. That was a forward contract. Centuries later, Tulip Mania showed the dark side: when a speculative boom in tulip futures burst, fortunes vanished. Derivatives have always served two purposes, protection and speculation.

1. Derivative = a contract whose value is derived from an underlying asset: metals, energy, agri commodities, or financial assets (shares, bonds, FX).
2. Forward: a bilateral OTC contract with customised terms; both parties are obliged to perform.
3. Futures: a standardised forward traded on an exchange (lot size, maturity and so on are fixed by the exchange), with settlement guaranteed by the clearing corporation.
4. Option: the right but not the obligation to buy (call) or sell (put). The buyer pays the premium; the writer receives it and carries the obligation.
5. Swap: an agreement to exchange future cash flows under a prearranged formula. In effect, a series of forward contracts.
6. Three participant types: hedgers (reduce risk), speculators/traders (take views and use leverage) and arbitrageurs (lock in risk-free profit from price differences).

7 OTC market: tailor-made contracts, decentralised counterparty risk, no formal centralised limits on positions, leverage or margining, and little disclosure.

8 Global milestones: CBOT 1848 (forwards); first exchange-traded futures 1865; CME's IMM currency futures 1972; CBOE in 1973 was the first listed options market; first stock index futures at Kansas City Board of Trade in 1982.

9 India: Dr L.C. Gupta committee (24 members, formed Nov 18, 1996; report Mar 17, 1998) recommended that derivatives be declared 'securities'. The Prof. J.R. Varma group (June 1998, report Oct 1998) set the risk containment framework.

10 SCRA was amended in 1999 to include derivatives in 'securities'. Index futures began in June 2000 (Nifty and Sensex), index options in June 2001, stock options in July 2001 and stock futures in November 2001. MSEI started in Feb 2013.

11 Significance: price discovery, transfer of risk from risk-averse to risk-seeking participants, and shifting speculation from unorganised to organised markets.

12 Risks: counterparty, price, liquidity, legal/regulatory and operational (fraud, poor documentation, improper execution). Read the Model Risk Disclosure Document.

L.C. Gupta Committee formed	Index futures start in India	Index options / Stock options	Stock futures
Nov 18, 1996 (24 members)	**June 2000**	**June 2001 / July 2001**	**November 2001**

SCRA amended to cover derivatives
1999

🎯 EXAM TIP
Expect sequence questions (which came first in India), the definitions of hedger, speculator and arbitrageur, and OTC versus exchange-traded features.

⚠️ TRAP
Swaps are a series of forwards, not of options. In India, stock options (July 2001) came before stock futures (November 2001).



Chapter 2 · Weight: 5%

Understanding the Index

An index is a portfolio that represents a market. The chapter covers how indices are weighted, what makes a good index, impact cost as a measure of liquidity, how indices are managed, and their uses: index funds, ETFs and index derivatives.

MAKE IT STICK
Think of the Nifty as a class average, where the percentage change matters more than the absolute level. In a price-weighted index, the student with the highest marks pulls the average most. In a market-cap index, the biggest company does. Under an equal-weight scheme, everyone counts the same and the teacher keeps rebalancing.

1. An index is a statistical indicator expressed as a change from a base value, so the percentage change matters more than the level.
2. Market-cap weighted: weight = shares × price. Index = current market cap / base market cap × base value.
3. Free-float market-cap index: only shares readily available for trading count. Sensex, Nifty and SX40 have moved to free float.
4. Price-weighted: index = sum of prices / number of stocks. Higher-priced stocks carry more weight (e.g. Dow Jones, Nikkei 225).
5. Equal-weighted: every stock gets the same weight. The manager must periodically sell the winners and buy the losers to rebalance.
6. A good index reflects market behaviour, is computed by an independent third party and is professionally maintained.
7. Bid-ask spread = transaction cost for a small trade. Impact cost = percentage degradation from the ideal price (the average of best bid and best ask) for a given order size.
8. Impact cost varies with order size, differs between the buy and sell sides, and is high when liquidity is poor.
9. Diversification: going from 10 to 20 stocks cuts risk sharply; beyond 100 stocks there is almost no further reduction.
10. BSE indices are managed by Asia Index Pvt Ltd and NSE indices by NSE Indices Ltd. An Index Committee decides inclusions and exclusions.
11. Maintenance = adjusting for corporate actions. Revision = changing the constituents.
12. SEBI (Oct 30, 2025) norms for derivatives on non-benchmark indices: at least 14 constituents, top weight ≤ 20%, top 3 combined ≤ 45%, descending weights. BANKEX and FINNIFTY were adjusted in a single tranche by Dec 31, 2025; BANKNIFTY in four monthly tranches by Mar 31, 2026.
13. Applications: index funds (small tracking error), index derivatives (hedging market risk) and ETFs (trade intraday like shares).

Ideal price	**(Best bid + Best ask) / 2**
Non-benchmark index: min constituents	**14**
Top constituent / Top-3 weight cap	**≤ 20% / ≤ 45%**
Price-weighted examples	**Dow Jones, Nikkei 225**

🎯 **EXAM TIP**
Expect impact cost calculations, price-weighted versus market-cap index arithmetic, and the new non-benchmark index weight caps.

⚠️ **TRAP**
Impact cost is not paid to anyone (it is not STT, brokerage or a fee). It arises from illiquidity and is LOW when liquidity is HIGH.

Chapter 3 · Weight: 20%

Introduction to Forwards and Futures

This is the futures engine room: how forwards fall short, how futures fix those problems, contract specifications, basis, cost of carry, margins and MTM, open interest, payoffs, pricing models, and how hedgers, speculators and arbitrageurs use futures.

MAKE IT STICK

You agree with your goldsmith today to buy 10 g of gold in a month at ₹62,337. That is a forward. If he vanishes, you face counterparty risk, and you cannot easily sell the deal to anyone else. Put that same promise on an exchange with a clearing corporation in the middle and daily MTM, and you have a futures contract.

1. Limitations of forwards: liquidity risk (tailor-made, hard to exit), counterparty/default/credit risk, lack of transparency and settlement complications.
2. In futures, the exchange fixes every term except the price. Both buyer and seller pay margins.
3. Index derivative contract value: at least ₹15 lakh at introduction, and lot size set so the value is ₹15–20 lakh on review day (SEBI Oct 1, 2024, effective Nov 2024). The earlier range was ₹5–10 lakh, set in 2015.
4. NSE futures follow a three-month cycle (near, next, far). NSE expiry is the last Tuesday of the month; BSE chose Thursday. Exchanges may choose only Tuesday or Thursday, and changing it needs SEBI approval.
5. Each exchange may offer weekly options on only one benchmark index. All other equity derivatives need a tenor of at least one month.
6. Daily settlement price = volume-weighted average of the last half hour. Final settlement price = closing price of the underlying in the cash market on expiry day.
7. Basis = Spot − Futures. It is negative when futures trade above spot (contango) and converges to zero at expiry.
8. Cost of carry for equity = interest cost − dividend income. Fair futures = Spot + Carry − Inflows, i.e. $F = S(1+r-q)^T$ or $F = S \cdot e^{((r-q)T)}$.
9. Futures above fair value leads to cash-and-carry arbitrage (buy cash, sell futures). Futures below fair value leads to reverse cash-and-carry (sell cash, buy futures).
10. Transaction costs, taxes and margins create no-arbitrage bounds around the fair price; lower costs narrow the bounds.
11. Cost of carry model assumptions: abundant supply, non-seasonal demand and supply, storable asset, short selling possible, no transaction costs, no taxes, no margins. Because it ignores margins, the model suits forwards better than futures.
12. Convenience yield (mainly for commodities under scarcity) can push futures below spot. Expectations model: futures = expected spot price. Futures > spot is a contango market; futures < spot is backwardation.

13 Open interest = contracts outstanding, counting one side only. Volume = contracts traded during the period.

14 Beta > 1 is aggressive and < 1 is conservative. Portfolio beta = Σ(weight × beta). Hedge contracts = (Portfolio value × β) / (futures level × lot size).

15 Long hedge protects a planned purchase; short hedge protects holdings you plan to sell. Cross hedge uses a related asset (e.g. crude for jet fuel). Hedge with the contract that expires just after the date you exit.

16 A calendar spread is measured against the near month and becomes a naked position when the near leg expires or either leg is closed. Only calendar spreads are available in equity F&O.

Index contract value (new) **₹15–20 lakh (min ₹15 lakh at launch)**	Daily settlement price **VWAP of last 30 minutes**	NSE / BSE expiry day **Last Tuesday / Last Thursday**
Sensex futures lot / cycle **20; 7 weekly + 3 monthly**	Trading hours **9:15 am – 3:30 pm**	Nifty futures tick **5 paise**

🎯 EXAM TIP
This chapter and Options together make up 40% of the paper. Practise fair-value calculations (both discrete and continuous), MTM, profit on long and short futures, hedge contract counts, portfolio beta, and spotting cash-and-carry versus reverse arbitrage.

⚠️ TRAP
In this workbook basis = SPOT – FUTURES, so futures at a premium means NEGATIVE basis. Also, round hedge contracts to the nearest whole number (13.22 → 13).



Chapter 4 · Weight: 20%

Introduction to Options

This chapter covers how options work: rights versus obligations, contract specifications, moneyness, intrinsic and time value, payoffs for the four basic positions, how options differ from futures, what drives premiums, the Greeks, pricing models and implied volatility.

MAKE IT STICK
An option is like an insurance policy. A put buyer pays a small premium and is covered if prices crash, much like car insurance after an accident. The writer pockets the premium every quiet month but must pay out on the bad day. That is why sellers post margin and buyers do not.

1 Buyer (holder) has the right and pays the premium upfront. Seller (writer) has the obligation and receives the premium.

2 American options can be exercised on or before expiry; European options only on expiry. All index and stock options in India are European.

3 Premium is quoted per unit; total premium = premium × lot size.

4 Moneyness for calls: ITM if S > X, ATM if S ≈ X (nearest strike), OTM if S < X. Puts are the reverse: ITM if S < X.

5 Intrinsic value: call = max(S – X, 0); put = max(X – S, 0). It is never negative. Time value = premium – intrinsic value. ATM and OTM options have only time value.

6 Long call: maximum loss is the premium, profit is unlimited, BEP = X + P. Short call: maximum profit is the premium, loss is unlimited, same BEP.

7 Long put: maximum loss is the premium, maximum profit = X – P (if the price falls to zero), BEP = X – P. Short put is the mirror image.

8 Option buyers pay no margin. Sellers pay margin and only they are subject to daily MTM margins. In futures, both sides pay margin and MTM.

9 Five price drivers: spot, strike, volatility, time to expiry and interest rate. A higher spot raises calls and lowers puts. A higher strike lowers calls and raises puts. Higher volatility or more time raises both. Higher interest rates raise calls and lower puts.

10 Time decay makes options 'wasting assets', which structurally favours sellers.

11 Delta = change in premium / change in underlying price. Long call 0 to +1, long put 0 to –1. Delta is also called the hedge ratio.

12 Gamma = change in delta per unit change in the underlying. Theta = time decay per day (negative for long options). Vega = sensitivity to a 1% change in volatility (positive for long calls and puts). Rho = sensitivity to the interest rate.

13 Binomial model: William Sharpe, 1978; iterative, accurate but complex. Black-Scholes: 1973; non-iterative, uses 5 inputs and ignores dividends.

14 Historical volatility is calculated from past prices. Implied volatility is backed out of the market premium by running Black-Scholes in reverse. Traders tend to sell when IV is high and buy when it is low.

15 Deep ITM options have the highest premium; ATM options carry the greatest uncertainty and therefore the most time value.

Call BEP / Put BEP	**X + P / X – P**
Option style in India	**European (index & stock)**
Sensex options cycle	**7 weekly, 3 monthly, 3 quarterly, 8 semi-annual**
Black-Scholes / Binomial	**1973 / 1978 (Sharpe)**
Index option tick	**5 paise**

🎯 **EXAM TIP**

Every mock and the real paper will ask for intrinsic value, time value, break-even, maximum loss or profit, and P&L at expiry for a given strike and premium (often times the lot size). Know the direction of each Greek.

⚠️ **TRAP**

The BEP is the same for a long and a short call (X + P). The BEP depends on the instrument, not on which side you are. Also, theta is negative for buyers, and vega is positive for long calls AND long puts.

Chapter 5 · Weight: 10%

Strategies using Equity Futures and Equity Options

This chapter puts the building blocks together: hedging with stock and index futures, leverage in speculation, cash-and-carry, reverse and calendar arbitrage, option spreads, straddles, strangles, covered calls, collars, butterflies, protective puts, put-call parity, delta hedging, and reading OI and PCR.

MAKE IT STICK

A long straddle is like betting on a cricket match ending with a big score by either team, without caring who wins. A short straddle is the groundsman betting on a rain-soaked draw. The butterfly is the cautious groundsman who also buys an umbrella on both sides.

1. Long hedge with stock futures locks in the price of a future purchase. Short hedge locks in the price of a planned sale. Either way the effective price lands close to the initial futures price.

2. Leverage example: buying shares returned 3.62% while a long futures position with a 20% margin returned 17.69% on a similar move.

3. Calendar spread arbitrage: compare the actual spread with the fair spread, then sell the relatively overpriced contract and buy the underpriced one. It is low risk and low return.

4. Vertical spread = same expiry, different strikes. Horizontal (calendar/time) spread = same strike, different expiry. Diagonal = different strike and expiry. Horizontal and diagonal spreads cannot be drawn on a payoff chart.

5. Bull call spread: buy the lower-strike call and sell the higher-strike call. Net debit = maximum loss; maximum profit = strike difference – debit; BEP = lower strike + debit.

6. Bull put spread: sell the higher-strike put and buy the lower-strike put. Net credit = maximum profit; maximum loss = difference – credit; BEP = higher strike – credit.

7. Bear call spread: sell the lower-strike call and buy the higher-strike call (net credit). Bear put spread: buy the higher-strike put and sell the lower-strike put (net debit).

8. Long straddle: buy a call and a put at the same strike and expiry. Maximum loss = total premium; BEPs = X ± total premium. It profits from big moves either way. A short straddle has limited profit and unlimited loss.

9. Long strangle: buy an OTM call and an OTM put (different strikes). It is cheaper; BEPs = put strike – total premium and call strike + total premium.

10. Covered call (long stock + short call) behaves like a synthetic short put. Protective put (long stock + long put) behaves like a synthetic long call.

11. Collar = covered call + long OTM put. It limits the downside but trims the upside.

12. Butterfly (calls): buy 1 low-strike, sell 2 middle-strike, buy 1 high-strike. Net debit = maximum loss; BEPs = low + debit and high – debit; limited profit and limited loss.

13 Put-call parity (European only): $c + X \cdot e^{(-rt)} = p + S_0$.

14 Delta hedge: futures delta ≈ 1. Short 10 lots of calls with delta 0.5 needs long 5 lots of futures. Rebalance as delta changes.

15 Rising price + rising OI = bullish (fresh longs). Rising price + falling OI = short covering. Falling price + rising OI = bearish (fresh shorts). Falling price + falling OI = long unwinding.

16 PCR = put OI (or volume) / call OI (or volume). It is a contrarian indicator: PCR < 1 is bearish and PCR > 1 is bullish.

17 OI is now measured as Future-Equivalent OI (delta-adjusted) per SEBI's May 29, 2025 circular.

Straddle BEPs	Butterfly BEPs
Strike ± (call premium + put premium)	**Lower strike + net debit; Upper strike − net debit**

Put-call parity	PCR < 1 signals	Futures delta
$c + X \cdot e^{(-rt)} = p + S$	**Bearish (contrarian reading)**	**≈ 1**

🎯 EXAM TIP

Strategy questions usually ask you to name the strategy from its legs, or to compute maximum profit, maximum loss or BEP. Build a 3-row payoff table (below the low strike, between strikes, above the high strike) and the answer drops out.

⚠️ TRAP

Both the bull put spread and the bear call spread start with a net CREDIT. Both the bull call spread and the bear put spread start with a net DEBIT. Don't flip them.



Chapter 6 · Weight: 10%

Trading Mechanism

This chapter covers how F&O trading actually happens: who the entities are, the user hierarchy, order types, price-time matching, operating ranges and dynamic price bands, stock and index eligibility, corporate-action adjustments, trading costs (STT and others), algo trading rules, data fields and the IRRA platform.

MAKE IT STICK

The trading screen is a well-run queue at a railway counter. The best price goes first, and among equal prices whoever came first gets served (price-time priority). The station master (corporate manager) sets limits for every counter (branch); each clerk (dealer) sees only his own tickets.

1. Trading Member (TM) trades for clients or on its own account. Trading-cum-Clearing Member clears its own trades and those of others. Professional Clearing Member is not a TM (typically a bank or custodian). Self-Clearing Member clears only its own and its clients' trades.

2. Sub-brokers ceased to exist from April 1, 2019 and migrated to become Authorised Persons or TMs.

3. SEBI permits equity derivatives trading hours between 9:00 am and 11:55 pm if risk systems allow; normal hours are 9:15 am to 3:30 pm.

4. User hierarchy: Corporate Manager (sets branch exposure limits), Branch Manager, Dealer (sees only own orders).

5. Time conditions: Day order and IOC (partial fill allowed, the rest cancelled immediately). Price conditions: limit, market and stop-loss (the trigger lies between the last traded price and the limit price).

6. Matching: price-time priority. The best buy is the highest bid and the best sell is the lowest ask.

7. There are no price bands in derivatives, only operating ranges: 10% of base price for index and stock futures, and a delta-based range for options.

8. Dynamic price band flexing (May 24, 2024): needs 50 trades, 10 unique traders and 3 brokers on each side. Flexes are 5% (after 15 minutes) twice, then 3% (after 30 minutes) twice, then 2% (after 60 minutes). Bands slide in both directions.

9. Stock eligibility (6 months continuously): top 500 by average market cap and traded value, MQSOS ≥ ₹75 lakh, MWPL ≥ ₹1,500 crore, average daily delivery value ≥ ₹35 crore.

10. If a stock fails for 3 consecutive months, no fresh month contracts are issued. Once excluded, it cannot be re-included for 1 year and must meet the criteria for 6 consecutive months.

11. Stock PSF exit: 15% of active TMs or 200 TMs (whichever is lower), trading on 75% of days, average daily turnover ≥ ₹75 crore, average daily notional OI ≥ ₹500 crore. Applies after 6 months from introduction.

12. Index eligibility: ≥ 80% of the index weight in F&O-eligible stocks, and no single ineligible stock above 5%. Index PSF: 15% of TMs or 20 TMs, 75% of days, turnover ₹10 crore, OI ₹4 crore. Re-inclusion after at least 6 months.

13. Corporate-action adjustments are made after close on the last cum-date. Bonus AF = (A+B)/B; split/consolidation AF = A/B. New strike = old strike / AF; new lot = old lot × AF.

14. Rights: E = (P − S)/(A + B); AF = (P − E)/P. Dividends above 2% of market value are extraordinary, and the full dividend is subtracted from strikes.

15. STT (from April 1, 2026): futures sale 0.05% (seller, on traded price); option sale 0.15% of premium (seller); exercised option 0.15% of settlement price (purchaser).

16. Stamp duty 0.002% on futures buys and 0.003% on options buys. SEBI fee ₹10/crore. NSE IPFT ₹10/crore (futures), ₹50/crore of premium (options). GST 18%.

17 Algo trading: ≤ 10 orders per second needs no registration (generic ID); > 10 OPS needs registration and a unique ID. Open APIs are not allowed and a static IP is required. Black-box algo providers must register as Research Analysts.

18 Glitches: inform the exchange within 1 hour, preliminary report by T+1, RCA within 14 days. IRRA lets investors only square off or cancel orders (not open new positions) when a broker's systems fail.

Futures operating range **10% of base price**	MQSOS / MWPL / Delivery value **≥ ₹75 lakh / ≥ ₹1,500 cr / ≥ ₹35 cr**	Ordinary vs extraordinary dividend **Above 2% of market value = extraordinary**
STT futures / options premium **0.05% / 0.15% (seller)**	Low-frequency algo threshold **≤ 10 orders per second**	Glitch RCA deadline **14 days**

🎯 EXAM TIP
Expect corporate-action adjustments (bonus, split, rights), STT calculations, order-type scenarios (especially stop-loss), stock and index eligibility numbers, and the algo rules.

⚠️ TRAP
For a bonus, AF = (A+B)/B, not A/B. A 1:1 bonus gives AF = 2, which halves the strike and doubles the lot. A split uses A/B directly.

Chapter 7 · Weight: 10%

Introduction to Clearing and Settlement System

This chapter covers what happens after the trade: clearing member types and net worth, computing open positions, interoperability, MTM and final settlement (cash for index, physical for stocks), SPAN margins, position limits, penalties, running account settlement, SGF and IPF, recent SEBI F&O measures and the cyber security framework.

MAKE IT STICK
The clearing corporation is the referee who steps between every buyer and seller (novation). Each evening it settles every player's daily score in cash (MTM) and holds a security deposit (margin) big enough to cover a bad day 99% of the time.

1 Novation: the clearing corporation becomes the central counterparty to every trade and guarantees settlement.

2 CM eligibility: net worth ₹300 lakh (₹100 lakh for a CM clearing only its own deals), ₹50 lakh security deposit, plus ₹10 lakh for each additional TM cleared.

3 Open position: proprietary positions are netted; client positions are grossed (each client's net, with longs and shorts added separately).

4 Interoperability (framework 2018, implemented 2019): a broker can clear trades from any exchange at one clearing corporation. Long 10 lots on NSE + short 6 on BSE = net long 4.

5 Since October 2019, stock F&O is settled by physical delivery while index F&O is settled in cash. Settlement is on T+1.

6 Daily MTM uses the last-30-minute VWAP across exchanges. For an untraded contract, theoretical F = S·e^(rt). MTM is cash-settled before the market opens the next day.

7 Physically settled stock options: the settlement value = strike × lot. ITM options are exercised automatically on expiry.

8 Net settlement of cash and F&O on expiry is available only to the same TM-CM combination and not to institutions.

9 SPAN (developed by CME) sets initial margin at a 99% one-day VaR, or two-day for futures where MTM cannot be collected before the next day.

10 Other margins: premium margin, assignment margin, exposure margin, ICMTM (futures only) and delivery margin, staggered 20%, 40%, 60% and 80% from expiry−4 to expiry−1 EOD.

11 Short Option Minimum Charge was removed (Feb 24, 2020). Net Option Value is added to liquid net worth. Client collateral may be given only through a margin pledge (from Sept 2020).

12 Peak margin: at least 4 snapshots a day. Since Aug 1, 2022, BOD margin parameters are used for the snapshots.

13 Client position limit in stock F&O: higher of 1% of free float or 5% of OI. Anyone holding ≥ 15% of OI in an index must disclose it.

14 TM limits: index futures and index options each higher of ₹7,500 crore or 15% of OI. Stock F&O: 20% of MWPL.

15 MWPL = 20% of free float (non-promoter shares). Alert at 60%; at > 95% no fresh positions from the next day (ban period); normal trading resumes at ≤ 80%.

16 PAN-level index options limit (FutEq): net EOD ₹1,500 crore, gross ₹10,000 crore. Intraday: net ₹5,000 crore, gross ₹10,000 crore, with at least 4 random snapshots (one between 14:45 and 15:30).

17 Index futures limits: FPI Cat I, MFs, TM prop and clients: higher of 15% of OI or ₹500 crore. FPI Cat II (others): 10%. FPI Cat II (individuals, family offices, corporates): 5%.

18 Running account: settled monthly or quarterly as the client chooses. Funds lying unused for 30 days are returned in the next monthly cycle.

19 SEBI measures: upfront option premium collection and no calendar-spread benefit on expiry day (both Feb 1, 2025); intraday position monitoring (Apr 1, 2025); an extra 2% ELM on short options on expiry day (Nov 20, 2024). For single-stock derivatives, no calendar-spread benefit on expiry day from May 5, 2026.

20 CSCRF has five goals: Anticipate, Withstand, Contain, Recover, Evolve. The MIIs are stock exchanges, depositories, clearing corporations, KRAs and QRTAs.

CM net worth / self-clearing	Initial margin VaR	MWPL
₹300 lakh / ₹100 lakh	**99%, 1-day (2-day for futures)**	**20% of free float shares**

MWPL ban / resume	Delivery margin staging	TM index futures limit
> 95% / ≤ 80%	20/40/60/80% on E-4…E-1	Higher of ₹7,500 cr or 15% of OI

🎯 EXAM TIP
Expect open-position (pro netted, clients grossed) puzzles, MTM and final settlement amounts, physical settlement values, margin types and MWPL thresholds.

⚠️ TRAP
Client positions are NOT netted across clients when computing a TM's open position or margins. Long 800 by C1 and short 900 by C2 gives an open position of 1,700, not 100.

Chapter 8 · Weight: 5%

Legal and Regulatory Environment

This chapter covers the laws behind F&O. SCRA 1956 defines securities and derivatives and makes exchange-traded derivatives legal (Section 18A). The SEBI Act 1992 sets SEBI's role. The chapter also covers L.C. Gupta committee norms, membership and net worth rules, clearing corporation duties, the default SOP and the SOP for exchange outages.

MAKE IT STICK
SCRA is the rulebook that says derivatives are a 'security'. SEBI is the referee. The L.C. Gupta committee designed the stadium: a separate governing council, at least 50 members, and certified players at the terminals.

1. SCRA Section 2(h) defines securities to include shares, derivatives, units of CIS and mutual funds, government securities, security receipts and rights or interests in securities, but not ULIPs.
2. Derivatives include contracts deriving value from the prices or index of prices of underlying securities, plus commodity derivatives.
3. Section 18A: derivative contracts are legal and valid if traded on a recognised stock exchange and settled through its clearing house.
4. SEBI's purpose: protect investors, develop the market and regulate the market.
5. SEBI accepted the L.C. Gupta recommendations on May 11, 1998, introducing derivatives in phases starting with index futures.
6. Derivatives exchange or segment: separate governing council, trading/clearing members limited to at most 40% of the council, at least 50 members, and prior SEBI approval for each contract.
7. CM net worth ₹300 lakh. Net worth = capital + free reserves minus non-allowable assets (fixed assets, pledged securities, member's card, unlisted securities, bad deliveries, doubtful debts, prepaid expenses, intangibles, 30% of marketable securities).
8. Membership: CM net worth ₹3 crore (with an auditor's certificate every 6 months), self-clearing member ₹1 crore, none prescribed by SEBI for TMs. Liquid net worth of at least ₹50 lakh. At least 2 certified approved users per TM.

9 Liquid assets must be at least 50% cash component (cash, BG, FDR, T-bills, G-secs).

10 Records such as trade confirmation slips and exercise notices are kept for 5 years. Every member is inspected at least once a year. A default in derivatives counts as a default on all segments and all exchanges.

11 The clearing corporation can transfer a defaulting member's client positions to another member. TGF guarantees bona fide trades, builds confidence and protects investors.

12 Default SOP: credit balances below ₹25 lakh are paid in full from unencumbered deposits; larger balances are paid pro rata.

13 Exchange outage SOP: inform SEBI immediately; inform participants within 15 minutes; give 15 minutes' notice before resuming. If trading does not resume at least 1 hour before close, or the outage hits in the last hour, all exchanges extend trading by 1.5 hours.

Section legalising derivatives	**SCRA Section 18A**
Min members of derivatives exchange	**50**
Broker representation on governing council	**Max 40%**
Records retention	**5 years**
Outage extension	**1.5 hours**

🎯 EXAM TIP
Expect Section 18A, the L.C. Gupta norms (50 members, 40% cap), net worth figures, TGF objectives (all of the above), and the timings in the default and outage SOPs.

⚠️ TRAP
SEBI prescribes net worth for CLEARING members; for TRADING members, net worth norms are set by the exchanges, not SEBI.

Chapter 9 · Weight: 5%

Accounting and Taxation

This chapter covers how futures and options appear in a client's books under ICAI guidance (margins, MTM, provisions on the prudence principle, final settlement) and how F&O income is taxed: as non-speculative business income, with presumptive taxation, audit thresholds and STT.

MAKE IT STICK
The accountant follows prudence: an expected loss is recorded today, while an expected gain waits until it is real. An open futures position showing an MTM loss gets a provision; an MTM gain stays out of P&L until the position is squared off.

1. AS-11 forwards for hedging: the premium or discount is amortised over the life of the contract. For trading forwards, no premium or discount is recognised and the gain or loss is marked at year end.

2. Initial margin is debited to 'Initial Margin – Equity Index/Stock Futures A/c' and shown under current assets. No entry is passed for the contract itself.

3. Margin given as a bank guarantee or securities needs only disclosure in the notes; no accounting entry is made.

4. MTM is paid or received through the 'Mark-to-Market Margin A/c'. At the balance sheet date, a debit balance gets a provision for loss charged to P&L. A credit balance (anticipated profit) is ignored and shown as a current liability.

5. On squaring up multiple open contracts of the same series, the weighted average method is used.

6. Options: the buyer debits 'Option Premium A/c'; the writer pays margin to 'Option Margin A/c'. At year end, the buyer provides for (premium paid – market premium) and the writer for (market premium – premium received).

7. F&O gains on a recognised exchange are non-speculative business income (Section 43(5) was amended by Finance Act 2005).

8. F&O losses can be set off against any income except salary, and carried forward for 8 assessment years against non-speculative business income, provided the return is filed on time. STT paid is deductible.

9. FPIs: F&O gains are taxed as capital gains, generally short-term.

10. Presumptive taxation under Section 44AD (turnover ≤ ₹2 crore) treats 6% of turnover as income.

11. Tax audit: mandatory if turnover exceeds ₹10 crore. For turnover of ₹2–10 crore, it applies when profits are below 6% without presumptive taxation; it is not required if more than 95% of transactions are digital.

12. STT: futures 0.05% (seller); option sale 0.15% of premium (seller); exercised option 0.15% of settlement price (purchaser). Physically settled contracts attract delivery STT of 0.1% on both sides.

Loss carry-forward **8 assessment years**	Presumptive income (44AD) **6% of turnover, turnover ≤ ₹2 cr**	Tax audit mandatory **Turnover > ₹10 crore**
Delivery STT on physical settlement **0.1% both sides**		

🎯 **EXAM TIP**

Expect the tax head (non-speculative business income), FPI treatment (STCG), the carry-forward period, the STT payer and rate, and which balance gets a provision.

⚠️ **TRAP**

F&O loss cannot be set off against SALARY income. MTM credit balances (profits) are never taken to P&L before settlement.

Chapter 10 · Weight: 5%

Sales Practices and Investors Protection Services

This chapter covers how to sell derivatives responsibly: client risk profiling, the Risk Disclosure Document and SEBI's F&O loss statistics, AML/KYC/CDD under PMLA, special-category clients, suspicious transaction reporting, UCC, and grievance redressal through exchanges, SCORES and the SMART ODR portal.

MAKE IT STICK

Selling F&O to a retired grandparent with low risk tolerance is like handing car keys to someone who has never driven. The rules require you to check their licence (KYC), explain the dangers (RDD), and report anything fishy (STR) without tipping them off.

1. Sales agents may call only between 9 am and 8 pm unless the customer requests otherwise, must identify themselves, and must stop on request. Churning (excessive trades for commission) is abuse.
2. Risk tolerance depends on age, income, family income, dependents, occupation, marital status, education, liabilities and inherited wealth.
3. The RDD must be signed at onboarding. Key risks are market, liquidity and counterparty (the last mostly absent on exchanges thanks to clearing guarantees).
4. SEBI study (Jan 2023, FY 2021–22): 9 out of 10 individual F&O traders lost money. The average loss was about ₹50,000. Loss makers spent an extra 28% of their losses on transaction costs; profit makers spent 15–50% of their profits on costs.
5. Brokers must show the risk disclosure pop-up at login and keep client P&L data for at least 5 years.
6. AML written procedures cover client acceptance, client identification, and transaction monitoring including STR.
7. CDD: identify the beneficial owner (the natural person with ultimate control), verify identity from independent sources, and continue scrutiny. Use a risk-based approach: enhanced CDD for high risk, simplified for low risk.
8. Clients of special category include NRIs, HNIs, trusts, charities and NGOs, PEPs, close family companies, clients from high-risk countries, non-face-to-face clients and clients with a dubious reputation.
9. PAN is mandatory for individuals. In-person verification is compulsory, except when Aadhaar-based e-KYC or online/DigiLocker KYC is used.
10. UCC is linked to PAN, uploaded to the exchanges and mapped to the demat account. The account is activated only after the UCC is generated.
11. STR goes to FIU-IND. Report internally to the Money Laundering Control Officer at once, never tip off the client, and keep dealing normally.
12. Digital KYC must be accessible to persons with disabilities, with alternate liveness checks for blind applicants.

13 Complaints: first the exchange or intermediary, then SEBI SCORES. The entity must resolve the complaint within 21 days. No fee is charged; anonymous or vague complaints are not entertained.

14 ODR: MII review within 21 days, conciliation within 21 days, then online arbitration. Payment due within 15 days of the award; intention to challenge must be filed within 7 days; if no stay within 3 months, the award must be complied with.

Sales call hours	Retail F&O loss makers	STR reported to	SCORES resolution time	ODR conciliation
9 am – 8 pm	**9 out of 10 (FY22)**	**FIU-IND**	**21 days**	**21 calendar days**

🎯 EXAM TIP

Expect PEP = client of special category, UCC, FIU-IND, SCORES, the RDD loss statistics and the ODR timelines.

⚠️ TRAP

After filing an STR, you must NOT inform the client and should normally continue dealing. Closing the account is only for exceptional cases.

🧮 Formulas

Contract value

```
Futures price × Lot size
```

Nifty futures at 24,000 with lot 75 → 24,000 × 75 = ₹18,00,000 (within the new ₹15–20 lakh band).

Futures P&L

```
Long: (Exit - Entry) × Lot; Short: (Entry - Exit) × Lot
```

Short 1 lot at ₹278, lot 1,200, buy back at ₹265 → (278 – 265) × 1,200 = +₹15,600.

Daily MTM

```
(Today's settlement price - Previous settlement price or trade price) × Lot × Number of lots
```

Long at 22,250, close 22,308.70, lot 25 → 58.70 × 25 = ₹1,467.50 (≈ ₹1,468) credited.

Basis

```
Basis = Spot price - Futures price
```

Spot 24,894.25, futures 25,006.60 → basis = –112.35 (negative, futures at premium).

Cost of carry (equity)

```
Carry = Interest cost - Dividend received; Fair F = S + Carry - Inflows
```

Share ₹100, borrow at 6% for 1 year, dividend ₹2 → 6 – 2 = ₹4; break-even futures = ₹104.

Fair futures price (discrete)

```
F = S × (1 + r - q)^T, T in years
```

Index 17,500, r = 12%, q = 4%, 90 days → 17,500 × 1.08^(90/365) = 17,835.26.

Fair futures price (continuous)

```
F = S × e^((r - q)T)
```

17,500 × e^(0.08 × 90/365) = 17,848.63. Futures above this → cash-and-carry arbitrage.

Cash-and-carry arbitrage profit

```
(Actual futures - Fair futures) × Lot
```

Stock ₹1,500, futures ₹1,550, carry 9% for 3 months: fair = 1,500 × e^(0.0225) = 1,534.13 → profit = 15.87 × 100 = ₹1,587.

Portfolio beta

$$\beta p = \Sigma (w_i \times \beta_i)$$

Betas 0.5, 1.1, 1.3, 0.9 with weights 35%, 15%, 20%, 30% → 0.175 + 0.165 + 0.26 + 0.27 = 0.87.

Number of index futures to hedge

```
N = (Portfolio value × βp) / (Futures index level × Lot size)
```

₹90,00,000 portfolio, β = 1.3, futures 17,700, lot 50 → 1,17,00,000 / 8,85,000 = 13.22 → short 13 contracts.

Intrinsic value

```
Call = max(S - X, 0); Put = max(X - S, 0)
```

Stock ₹286, 260 call → IV = 26. Put 18,400 with Nifty 18,315.10 → IV = 84.90.

Time value

```
Premium - Intrinsic value
```

Put premium 177.60 – IV 84.90 = ₹92.70 time value.

Break-even (single option)

```
Call BEP = X + P; Put BEP = X - P
```

17,500 call at ₹95 → BEP 17,595. 17,500 put at ₹150 → BEP 17,350.

Maximum loss of short put

```
Strike - Premium (if the price falls to zero)
```

Sell 245 put for ₹49 → maximum loss = 245 – 49 = ₹196 per share.

Option ROI

```
Net profit / Premium paid
```

17,600 call at ₹130, index ends 17,800 → profit 70 → ROI = 70/130 = 54%.

Bull call spread

```
Net debit = P(low) - P(high); Max loss = net debit; Max profit = (X₂ - X₁) - debit; BEP = X₁ + debit
```

Buy 17,500 call @185, sell 17,800 call @61 → debit 124, max profit 176, BEP 17,624.

Bull put spread

```
Net credit = P(high) - P(low); Max profit = credit; Max loss = (X₂ - X₁) - credit; BEP = X₂ - credit
```

Sell 17,500 put @125, buy 17,000 put @34 → credit 91, max loss 409, BEP 17,409.

Bear call / bear put spread

```
Bear call: credit = max profit, max loss = diff - credit, BEP = X₁ + credit. Bear put: debit = max loss, max profit = diff - debit, BEP = X₂ - debit
```

Bear put: buy 17,500 put @125, sell 17,000 put @34 → max loss 91, max profit 409, BEP 17,409.

Straddle

```
Total premium = Pc + Pp; BEPs = X ± total premium
```

6,000 call @257 + 6,000 put @136 → max loss (long) 393; BEPs 5,607 and 6,393.

Strangle

```
BEP low = Xput - total premium; BEP high = Xcall + total premium
```

Buy 6,200 call @145 + 6,000 put @140 → total 285; BEPs 5,715 and 6,485.

Butterfly (calls)

```
Net debit = P₁ + P₃ - 2P₂; Max profit = (X₂ - X₁) - debit; BEPs = X₁ + debit, X₃ - debit
```

6,000 @230, 2 × 6,100 @150, 6,200 @100 → debit 30; max profit 70; BEPs 6,030 and 6,170.

Covered call

```
BEP = Stock cost - call premium; Max profit = (X - Stock cost) + premium
```

Buy stock 1,590, sell 1,600 call @10 → BEP 1,580; max profit 20.

Protective put

```
Max loss = (Stock cost - X) + put premium
```

Stock 1,600 + 1,600 put @20 → max loss ₹20; price 1,660 → net +40.

Put-call parity

```
c + X·e^(-rt) = p + S₀
```

S = 1,251, X = 1,240, c = 47.50, r = 8%, t = 1/12 → p = 47.50 + 1,240e^(−0.08/12) − 1,251 = 28.26.

Delta, gamma, theta

```
Δ = ΔPremium/ΔS; new Δ ≈ Δ + Γ × ΔS; ΔPremium ≈ θ per day
```

Delta 0.60: stock +₹1 → call +₹0.60. Delta 0.50, gamma 0.08: stock +₹1 → new delta 0.58.

Delta-neutral hedge

```
Futures lots = |option delta × option lots| (same lot size); calls to write for n shares = n / delta
```

Short 10 lots of calls, delta 0.5 → buy 5 lots of futures. If delta rises to 0.6 → 6 lots.

Put-call ratio

```
PCR = Put OI (or volume) / Call OI (or volume)
```

Put OI 8,000 / Call OI 12,000 = 0.67 → bearish signal (contrarian).

Impact cost

```
(Actual execution price - Ideal price) / Ideal price × 100; Ideal = (best bid + best ask)/2
```

Bid 9.80/ask 9.90 → ideal 9.85; buy 1,500 at avg 9.9333 → impact cost 0.84%.

Index value

```
Market-cap index = Current Mcap / Base Mcap × Base; Price index = Σ prices / n
```

Mcap 18,800 → 42,500 lakh on base 100 → 226.06. Price index (150+300+450+100+250)/5 = 250.

Bonus / split adjustment

```
Bonus A:B → AF = (A+B)/B; Split A:B → AF = A/B; New strike = Old/AF; New lot = Old × AF
```

1:1 bonus → AF 2: strike 1,000 → 500, lot 500 → 1,000.

Rights adjustment

```
E = (P - S)/(A + B); AF = (P - E)/P; New strike = Old × AF
```

P = 500, S = 300, 1 for 4 → E = 200/5 = 40; AF = 460/500 = 0.92.

Futures settlement (theoretical, untraded)

```
F = S × e^(rt)
```

S = 1,000, r = 8%, t = 0.25 → F = 1,020.20.

STT

```
Futures: 0.05% × traded value (seller); Options: 0.15% × premium (seller); Exercised: 0.15% × settlement value (buyer)
```

Short 1 lot Nifty futures 17,500 × 50 = ₹8,75,000 → STT 0.05% = ₹437.50.

Leverage ROI

```
ROI on futures = Profit / Margin
```

Profit ₹39,100 on margin ₹2,21,000 = 17.69% vs 3.62% on outright share purchase.

🔢 Numbers to remember

100 Qs × 1 mark, 2 hours, pass 60%, −25% negative
Exam

Ch3 & Ch4 20% each; Ch1, 5, 6, 7 10% each; Ch2, 8, 9, 10 5% each
Syllabus weights

₹15–20 lakh (min ₹15 lakh at launch), from Nov 20, 2024
Index contract value

₹5–10 lakh (set in 2015)
Old contract value

Only one benchmark index per exchange
Weekly expiries

NSE Tuesday, BSE Thursday
Expiry days

20
Sensex lot size

VWAP of last 30 minutes
Daily settlement price

10% of base price
Futures operating range

50 trades, 10 unique traders, 3 brokers per side
Price-band flex trigger

Top 500; MQSOS ≥ ₹75 lakh; MWPL ≥ ₹1,500 cr; delivery ≥ ₹35 cr; 6 months
Stock F&O eligibility

Fails 3 months → no new contracts; re-inclusion after 1 year
Stock exit / re-entry

15% of TMs or 200; 75% days; ₹75 cr turnover; ₹500 cr OI
Stock PSF

15% of TMs or 20; 75% days; ₹10 cr turnover; ₹4 cr OI
Index PSF

≥ 80% weight eligible; no ineligible stock > 5%
Index eligibility

≥ 80% constituents, ≥ 50% overlap, correlation ≥ 0.90 (6 months)
Surrogate index

> 2% of market value
Extraordinary dividend

Futures 0.05%; options 0.15% of premium; exercise 0.15% of settlement price
STT

Futures 0.002%; options 0.003%
Stamp duty (buyer)

₹10 per crore
SEBI turnover fee

₹300 lakh (₹3 cr); self-clearing ₹100 lakh (₹1 cr)
CM net worth

₹50 lakh + ₹10 lakh per extra TM
CM deposits

At least ₹50 lakh
Liquid net worth

99% VaR, 1-day (2-day for futures)
Initial margin

20%, 40%, 60%, 80% from E-4 to E-1
Delivery margin

20% of free float; alert 60%; ban > 95%; resume ≤ 80%
MWPL

Higher of 1% free float or 5% of OI
Client stock limit

Higher of ₹7,500 cr or 15% of OI
TM index limit

20% of MWPL
TM stock limit

≥ 15% of OI
Index OI disclosure

Net EOD ₹1,500 cr; gross ₹10,000 cr; intraday net ₹5,000 cr
PAN index options limit

2% on short options
Extra ELM on expiry day

> 10 orders per second
Algo registration

Inform 1 hr; prelim T+1; RCA 14 days
Glitch timelines

≥ 50 members; brokers ≤ 40% of council
Derivatives exchange

Small investors < ₹25 lakh paid first
Default SOP

8 assessment years
Loss carry forward

6% of turnover (≤ ₹2 cr)
Presumptive tax

21 days; ODR award payment 15 days
SCORES / ODR

9/10 traders lost; avg loss ≈ ₹50,000
SEBI study

📖 Glossary

Underlying	The asset (stock, index, commodity) from which a derivative derives its value.
Forward	Customised bilateral OTC agreement to buy or sell at a future date at a price fixed today.
Futures	Standardised, exchange-traded forward with margins, daily MTM and clearing corporation guarantee.
Option premium	Price paid by the option buyer to the writer for the right.
Strike (exercise) price	Price at which the option holder may buy (call) or sell (put).
European option	Exercisable only on expiry. All Indian index and stock options are European.
Moneyness	Whether an option is ITM, ATM or OTM, by comparing spot with strike.
Intrinsic value	Amount by which an option is in the money; never negative.
Time value	Premium minus intrinsic value; decays to zero by expiry.
Basis	Spot price minus futures price; becomes zero at expiry.
Cost of carry	Financing (and storage) cost minus income from holding the asset until delivery.
Contango / Backwardation	Futures above spot (expected rise) / futures below spot (expected fall).
Convenience yield	Intangible benefit of physically holding a scarce commodity; can push futures below spot.
Mark-to-market (MTM)	Daily settlement of futures profits and losses against the daily settlement price.
Open interest	Number of contracts outstanding (one side counted).
FutEq OI	Future-equivalent OI: positions multiplied by delta, now used for limits.
Beta	Sensitivity of a stock or portfolio to index movements; measure of systematic risk.
Hedge ratio	Number of hedge units per unit of exposure. For options, delta is the hedge ratio.
Cash-and-carry arbitrage	Buy spot, sell overpriced futures, carry to expiry.
Reverse cash-and-carry	Sell spot (or borrow and sell), buy underpriced futures.
Calendar spread	Long and short futures (or options) on the same underlying with different expiries.
Implied volatility	Volatility backed out of the market option price using a pricing model.
Delta / Gamma / Theta / Vega / Rho	Sensitivity of the premium to price / of delta to price / to time / to volatility / to interest rate.

Novation	Clearing corporation substitutes itself as counterparty to both sides of every trade.
SPAN	Standard Portfolio Analysis of Risk (CME): portfolio-based 1-day worst-case initial margin.
MWPL	Market-wide position limit: 20% of a stock's free-float shares.
Impact cost	Percentage price degradation from the ideal price when executing an order of a given size.
MQSOS	Median quarter-sigma order size: a liquidity test for F&O stock eligibility (≥ ₹75 lakh).
Adjustment factor	Ratio used to change strike, lot and position after corporate actions.
IRRA	Investor Risk Reduction Access: exchange platform to square off or cancel orders when a broker's system fails.
UCC	Unique Client Code linked to PAN; identifies a client on exchanges.
STR	Suspicious Transaction Report filed with FIU-IND under PMLA.
SCORES	SEBI's web-based centralised complaint redressal system.
PCR	Put-call ratio of OI or volume; a contrarian sentiment indicator.

⚡ Quick revision: the night before

1. Exam: 100 Qs, 100 marks, 120 minutes, 60% to pass, −0.25 per wrong answer.
2. Futures = standardised exchange-traded forwards; the exchange fixes everything except price.
3. Index futures started in India in June 2000; stock futures came last, in Nov 2001.
4. Basis = Spot − Futures; it converges to zero at expiry.
5. Fair F = S(1 + r − q)^T or S·e^((r−q)T). Equity carry = interest − dividend.
6. Futures > fair → cash-and-carry (buy spot, sell futures). Futures < fair → reverse.
7. Hedge lots = Portfolio × β ÷ (futures level × lot). Round off.
8. Long hedge protects a future purchase; short hedge protects holdings you will sell.
9. Pick the hedge contract expiring just AFTER your exit date.
10. Call IV = S − X; put IV = X − S; never negative. TV = premium − IV.
11. Call BEP = X + P, put BEP = X − P. Same BEP for long and short of the same option.
12. Option buyer: maximum loss = premium, no margin. Writer: margin plus MTM.
13. Volatility up or time up → both calls and puts rise. Rates up → calls up, puts down.
14. Delta: call 0 to 1, put −1 to 0. Gamma = change in delta. Theta negative for buyers. Vega positive for buyers.
15. Black-Scholes 1973; Binomial (Sharpe) 1978.
16. Bull call and bear put = net debit. Bull put and bear call = net credit.
17. Straddle BEPs = X ± total premium; strangle uses two different OTM strikes.
18. Covered call ≈ synthetic short put; protective put ≈ synthetic long call.
19. Butterfly: +1 low, −2 middle, +1 high; limited risk and limited reward.
20. Put-call parity: c + X·e^(−rt) = p + S (European only).
21. Price ↑ OI ↑ bullish; price ↑ OI ↓ short covering; price ↓ OI ↑ bearish; price ↓ OI ↓ long unwinding.
22. PCR < 1 bearish, > 1 bullish (contrarian).
23. Index contract value ₹15–20 lakh; one weekly benchmark index per exchange.
24. NSE expiry: last Tuesday; BSE: Thursday. Daily settlement price = last-30-minute VWAP.
25. Stock F&O physically settled since Oct 2019; index F&O cash-settled; settlement on T+1.
26. Bonus AF = (A+B)/B; split AF = A/B; strike ÷ AF, lot × AF.
27. Dividend above 2% of price = extraordinary → strikes reduced by the full dividend.
28. STT: futures 0.05%, options 0.15% of premium (seller), exercise 0.15% (buyer).
29. Initial margin = 99% one-day VaR via SPAN (two-day for futures).
30. MWPL = 20% of free float; ban above 95%, trading resumes at 80%.
31. Client stock limit = higher of 1% free float or 5% OI; TM index limit = higher of ₹7,500 cr or 15% OI.
32. Pro positions are netted; client positions are grossed for margin.
33. CM net worth ₹3 cr, SCM ₹1 cr, liquid net worth ₹50 lakh, 2 approved users per TM.
34. SCRA Section 18A makes exchange-traded derivatives legal.
35. Derivatives exchange: ≥ 50 members, brokers ≤ 40% of governing council.
36. F&O income = non-speculative business income; loss carried forward 8 years; no set-off against salary.
37. FPIs: F&O gains are capital gains (mostly short term).
38. STR → FIU-IND; never tip off the client.
39. SCORES: entity must resolve within 21 days; ODR conciliation 21 days.
40. 9 out of 10 individual F&O traders lost money (SEBI FY22 study).

🗓️ 21-day study plan (2 h/day)

Days 1–2: Ch 1–2: Derivative basics and indices

- Read Ch 1 and 2 notes; build a timeline of India's F&O launches
- Solve index (market-cap, price-weighted) and impact cost examples by hand
- Take Mock 1 untimed and review every explanation

Days 3–7: Ch 3: Futures (20%)

- Master the basis, cost of carry and fair-value formulas, both discrete and continuous, in Excel
- Drill MTM, futures P&L, hedge-contract and portfolio-beta numericals (20 a day)
- Learn the contract specification facts: lot value band, expiry days, settlement prices
- Take Mock 2

Days 8–12: Ch 4: Options (20%)

- Draw the four basic payoff charts from memory
- Practise IV, TV, BEP and max-loss questions; check them with a payoff table
- Make a Greeks direction table (sign for long call, long put, short call, short put)
- Take Mock 3

Days 13–15: Ch 5: Strategies

- Build an Excel payoff sheet for spreads, straddles, strangles, butterflies, collars
- Practise put-call parity and delta-hedge calculations
- Memorise the OI/price matrix and PCR interpretation
- Take Mock 4

Days 16–18: Ch 6–7: Trading, clearing, margins, limits

- Flashcards for eligibility criteria, MWPL, position limits and margin types
- Practise corporate-action adjustments and STT computations
- Solve open-position (pro netted vs client gross) puzzles
- Take Mock 5

Days 19–20: Ch 8–10 plus tough mocks

- Revise SCRA, membership norms, accounting entries, tax rules, KYC/AML, SCORES/ODR
- Take Mock 6 under timed conditions (36 minutes)
- Take Mock 7 under timed conditions and log every error by chapter

Days 21: Final revision

- Read Quick Revision, Numbers to Remember and Formulas twice
- Run the 100-question simulator once
- Light review only; sleep well

Independent study notes prepared from the official NISM workbook. Verify against the latest workbook on nism.ac.in.